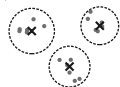
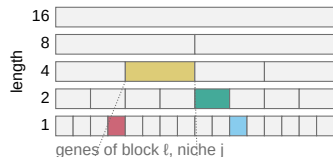


A

1 Index dyadic intervals

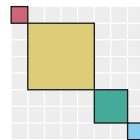


2 Split the query into dyadic pieces

contiguous: genes 3–10 $\rightarrow$ 4 piecesnon-contiguous: genes 1–2, 9–13 $\rightarrow$ 5 pieces

Each run of consecutive genes (in the block's order) gives at most $2 \log_2(\text{run length})$ pieces. Repeated in every niche's gene order.

3 Score each piece

query covariance on S

colour: piece blocks Q_i , compared with the cell's centroids

grey: used only in the exact re-rank

$$d_i(t) = \|\log Q_i - C_i(t)\|_F / \sqrt{p_i}$$

$C_i(t)$: centroid of tile t 's cluster in the cell of piece i ; p_i genes

4 Combine, re-rank

$$D(t)^2 = \sum_i p_i d_i(t)^2 / |S|$$

